MAPS FROM PERFECT COMPLEXES WITH SUPPORT LIFT FROM AN OPEN TO THE AMBIENT SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Shift lift with support. lemma-lift-perfect-complex-plus-shift-support (carded L26775). Provides Q on X supported on T with Q|_{U cap V} = P + P[1].
- Map lifting by tensoring with perfect complexes. lemma-lift-map (same chapter, carded). Replaces Q by Q tensor^L I to lift the composed map to E|_V.
- Gluing morphisms of complexes. Cohomology, lemma glue (cross-chapter, named). Assembles a: R -> E from the two local pieces.

Source reference. Original reference 08DG: focused-seeds/geometry/S9-perf2-lift-map-from-perfect-complex-with-support/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08EH: sources/stacks-perfect/pinned-original-source.tex, source lines 3571–3571. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08EK: sources/stacks-perfect/pinned-original-source.tex, source lines 3689–3689. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme. Let $U \subset X$ be a quasi-compact open. Let $T \subset X$ be a closed subset with $X \setminus T$ retro-compact in X. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Let $\alpha : P \to E|_U$ be a map where P is a perfect object of $D(\mathcal{O}_U)$ supported on $T \cap U$. Then there exists a map $\beta : R \to E$ where R is a perfect object of $D(\mathcal{O}_X)$ supported on T such that P is a direct summand of $R|_U$ in $D(\mathcal{O}_U)$ compatible α and $\beta|_U$.*

Proof. Since X is quasi-compact there exists an integer m such that $X = U \cup V_1 \cup \ldots \cup V_m$ for some affine opens V_j of X. Arguing by induction on m we see that we may assume $m = 1$. In other words, we may assume that $X = U \cup V$ with V affine. By Lemma 08EK we can choose a perfect object Q in $D(\mathcal{O}_V)$ supported on $T \cap V$

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and an isomorphism $Q|_{U\cap V} \to (P \oplus P[1])|_{U\cap V}$. By Lemma 08EH we can replace Q by $Q \otimes^{\mathbf{L}} I$ (still supported on $T \cap V$) and assume that the map

$$Q|_{U\cap V} \to (P \oplus P[1])|_{U\cap V} \longrightarrow P|_{U\cap V} \longrightarrow E|_{U\cap V}$$

lifts to $Q \to E|_V$. By Cohomology, Lemma 08DG we find an morphism $a : R \to E$ of $D(\mathcal{O}_X)$ such that $a|_U$ is isomorphic to $P \oplus P[1] \to E|_U$ and $a|_V$ isomorphic to $Q \to E|_V$. Thus R is perfect and supported on T as desired. $\square$